Research Plan: Guided Waves in a Smart Semiconductor Plate

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Problem statement

Consider a thin plate of thickness 2h made of a piezoelectric semiconductor. Both faces of the plate are free. The material is elastic, piezoelectric and thermally conducting, and it contains mobile charge carriers. Heat conduction follows the Moore–Gibson–Thompson (MGT) model, and size effects are described by the nonlocal strain gradient theory, which has two length scales.

Study the Lamb waves that travel along this plate. For the symmetric and antisymmetric modes, find the phase velocity and the attenuation as functions of frequency and plate thickness. Show how they change with the nonlocal length, the gradient length, the thermal relaxation time and the carrier concentration, and compare open-circuit and short-circuit faces.

Understanding the problem

This section explains every term used in the problem statement, one by one. If a word in a paper looks unfamiliar later, come back here first.

Why a plate? Most of our earlier papers treat a half-space, a block of material that goes on forever below its surface. Real sensors, filters and resonators use thin plates or films, and a plate behaves differently because the wave keeps bouncing between its two faces. That difference is the reason for this study.

Wave and plate

A wave is a disturbance that moves through a material and carries energy with it. Shake one end of a long rope and the bump runs along the rope, while the rope itself stays where it is. A plate is a flat body whose thickness is much smaller than its length and width, like a steel ruler or a sheet of tin.

Lamb wave

In a plate, a wave keeps reflecting from the top and bottom faces. The reflected parts add up and form a pattern that moves along the plate. This guided wave is called a Lamb wave, after Horace Lamb, who described it in 1917. The ringing you hear when you tap a thin metal sheet is the whole sheet vibrating, not only its surface.

Frequency, wavenumber and speed

The angular frequency ω is 2π times the number of vibrations per second. The wavenumber k is 2π divided by the wavelength. The phase velocity is $c = \omega/k$, which is the speed at which a crest moves along the plate.

Dispersion

In a plate, waves of different frequencies travel at different speeds. This is called dispersion. A glass prism shows the same idea with light: each colour moves at a slightly different speed in glass, so white light spreads into a rainbow. A dispersion curve is a graph of frequency against wavenumber, and most of our figures are built from it.

Modes

At a given frequency, a plate can carry a wave in more than one pattern, and each pattern is called a mode. In the usual case, symmetric modes stretch and squeeze the plate along its length, and antisymmetric modes bend it like a flapping sheet. They are named S0, S1, S2 and so on, and A0, A1, A2 and so on. In our piezoelectric plate, whether the modes split cleanly into these two families depends on the polarisation direction. We will fix that direction in step 2.

Cutoff frequency

Higher modes do not exist at every frequency. Each one appears only above its own cutoff frequency. A road is a rough picture: a narrow road carries one line of vehicles and a wider road carries several. At low frequency the plate carries only the first modes, and more become possible as the frequency rises.

Piezoelectricity

Some crystals produce a voltage when they are squeezed, and they change shape when a voltage is applied to them. The spark in a gas lighter comes from pressing such a crystal, and a quartz watch uses the opposite effect. In our plate, the stretching of the wave

creates an electric potential that travels along with it.

Heat and the MGT model

Stretching a material slightly changes its temperature, and a change in temperature makes it expand or shrink. Stretch a rubber band quickly and touch it to your lip, and it feels warm. So elastic waves and heat are linked, and that is why the problem is called thermoelastic.

The classical law of heat conduction, Fourier's law, says that a temperature change is felt everywhere at once. That cannot be exactly true. Put a pan on a stove and the handle takes a while to warm up. The Moore–Gibson–Thompson (MGT) model fixes this by letting heat travel at a finite speed, using a thermal relaxation time. Our recent papers already use it, so the equations are familiar.

Semiconductor and mobile charges

A semiconductor has free charge carriers, electrons or holes, that can move through the material. How many there are depends on how the material is made. ZnO and GaAs are both piezoelectric and semiconducting. The electric field of the wave pushes the carriers. They move and partly cancel the field, which is called screening, and some energy is lost in the process, so the wave changes speed and becomes weaker. Picture a crowded hall: if one corner is pushed, people move and the crowding evens out.

Size effect

Classical elasticity says that the stress at a point depends only on the strain at that same point. That works very well for everyday objects. At micro and nano scales, tests on tiny beams and plates show behaviour that this rule cannot explain.

Nonlocal theory, started by Eringen, says the stress at a point depends on the strain in a neighbourhood around it. Strain gradient theory says it also depends on how quickly the strain changes from place to place. The nonlocal strain gradient theory uses both ideas and has two length scales, one for each. When the plate is much thicker than these lengths, the classical answer comes back. Think of a person in a crowded lift: what they feel depends on the people around them, not only on the spot where they stand.

Attenuation and complex wavenumber

If a wave loses energy as it travels, the wavenumber is no longer a real number. It is written k = k_r + i k_i. The real part k_r gives the wavelength and the speed. The imaginary part k_i tells how fast the amplitude falls, in proportion to e^(−k_i x). A struck bell is the usual picture: the sound is loud at first and then slowly fades. In our plate, heat flow and mobile carriers are the main causes of fading. When the figures mention complex branches, they mean curves where k has both a real and an imaginary part.

The model we will use

The model is kept small so that each part can be checked on its own.

The plate has thickness 2h, and the wave travels along its length. We work in plane strain, which means the motion takes place in the length and thickness directions only. The top and bottom faces carry no stress. They can be thermally insulated or held at a fixed temperature, and electrically open or short-circuited, and no carriers flow out through them.

Inside the plate we use the elastic equations with piezoelectric coupling, the MGT heat equation, and the linearised carrier equation. These are the same forms as in our recent piezo-semiconductor papers, so the algebra is already familiar. The size effect enters through the nonlocal strain gradient law with two length scales.

We leave out surface elasticity, flexoelectricity, memory-dependent derivatives, graded materials, nonlinear effects and full 3D geometry. Each of these can be a separate paper later.

Two points must be written clearly in the paper. First, the size-effect theory asks for extra boundary conditions on the faces. We apply the usual ones, say so openly, and show how much the results change when the length scales change. Second, the polarisation direction is still open. We fix it in step 2, because it decides whether the modes split into symmetric and antisymmetric families.

Work plan

The work is done in six steps. After each step there is a check against a known result. Please send me that check plot before you start the next step, so that if something is wrong we know which step caused it.

Step	What you do	The check before moving on	What you gain
1. Plain elastic plate	Derive and solve the classical Rayleigh–Lamb problem	Known cutoff frequencies and thin-plate limits match; for a very thick plate the speed approaches our Rayleigh half-space result	Confidence with the method
2. Add piezoelectricity	Include the electric potential, for open and short-circuited faces	Switching the coupling off returns step 1; the open/short speed difference gives the coupling coefficient K^2	A first result worth showing
3. Add heat (MGT)	Include the temperature field and the MGT heat equation	Switching thermal coupling off returns step 2; the classical thermoelastic limit agrees with published results	A thermal effect on speed and fading
4. Add semiconductor carriers	Include the carrier concentration	Setting carrier density to zero returns step 3	The device-relevant physics
5. Add the size effect	Include the nonlocal strain gradient terms	Setting both length scales to zero returns step 4; nonlocal-only and gradient-only cases are checked separately	The main new contribution
6. Analysis	Track complex branches, fading, maps and sensitivity	Branches vary smoothly and root counts stay consistent along each branch	The figures that carry the paper

After step 3 we already have a verified piezo-thermal plate, which is a usable base for the paper. Steps 4 and 5 add the new parts. If step 5 turns out to be heavy, we will decide together what to do.

Calculations and figures

For each wavenumber we find the complex frequencies at which the plate can carry a wave, and then follow each branch from one wavenumber to the next so that curves do not get mixed up. I will give you working code for this part, so your time goes into understanding the results and making the figures.

Figure	What it shows	Why the paper needs it
Dispersion curves	Frequency against wavenumber for each mode	The base result
Complex branches	Real and imaginary parts together	Sets the paper apart from half-space work
Phase speed against frequency–thickness	Where each mode switches on and how it settles	Shows true plate behaviour
Fading against frequency	Effect of heat and carriers on how quickly waves die out	Links to device performance
Thickness–frequency–speed surface	The whole parameter landscape in one view	Rich information in one figure
Length-scale contour maps	Separate effects of the two size parameters	The main result of the size model
Coupling coefficient K^2 against thickness	Open versus short-circuit difference	Useful for sensor design
Mode shapes	Displacement, potential, temperature and carriers across the thickness	Builds physical understanding
Sensitivity ranking	Which parameter changes the speed most	Shows how far results can be trusted
Verification table	Each limiting case and its match	Builds reviewers' trust

All figures use one colour scheme, one font size and a resolution of 300 dpi, so the paper looks like one consistent piece of work.

Possible difficulties

These are the problems I expect along the way. None of them is unusual.

Hurdle	What it looks like	How we handle it
Extra boundary conditions in the size-effect theory	A reviewer may ask why only the usual conditions are applied	We state our choice openly and show how little or how much the results move when the length scales change
A missing or tangled branch	A curve jumps or two curves swap places	Follow each branch step by step and check the root count along the way
Uncertain parameter values	Length scales and carrier data vary between sources	Use ranges from the literature, never call a value measured, and show the sensitivity
Doubt about novelty	Someone may already have published something close	Read the closest papers fully before we write any claim
A reviewer who dislikes theory mixing	They may ask why all these effects sit together	Lead with the plate sensor motivation and the verification table
Too much at once	The full model can feel large	Work step by step, and treat step 5 as the last addition

If you get stuck at any step, send me the plot that looks wrong and we will go through it together.

Reading list

Eight items are enough to start. Read the first two first: they show the structure of the model and what complex branches look like. Read for the idea first and the algebra afterwards.

#	Paper	Why read it
1	Ray, "Dispersion curves for Rayleigh–Lamb waves in a micro-plate considering strain gradient elasticity", Wave Motion 86, 2019	The closest model for a plate with a size effect; shows how symmetric and antisymmetric dispersion is built
2	Dhib et al., "Real, imaginary and complex branches of Lamb waves in p-type piezoelectric semiconductor GaAs plate", Mater. Sci. Semicond. Process. 183, 2024	Shows complex branches in a semiconductor plate and how mobile charges screen the electric field, with experiment
3	Lim, Zhang and Reddy, "A higher-order nonlocal elasticity and strain gradient theory and its applications in wave propagation", J. Mech. Phys. Solids 78, 2015	The source of the nonlocal strain gradient theory we use; confirm the exact citation before citing
4	Our own papers: Gupta et al., "Piezo-semiconductor-void interplay in the size-dependent dynamics of Rayleigh waves within the Moore–Gibson–Thompson framework", IJNMHFF 35(9), 2025, and Gupta et al., "Vibrational analysis of size-dependent thermo-piezo-photo-electric semiconductor medium under memory-dependent MGT photo-thermoelasticity", MAMS 31(28), 2024	The governing equations we will reuse in the plate
5	Rayleigh wave in an elastic half-space with an attached piezoelectric semiconductor layer, with flexoelectricity and size effects, Appl. Math. Mech., 2025	A recent layered semiconductor wave paper; shows the device motivation and a numerical approach
6	Shaat, "Infeasibility of the nonlocal strain gradient theory for applied Physics", arXiv, 2017	A critique of the theory; good to know so we can answer it if a reviewer raises it

#	Paper	Why read it
7	Size-dependent vibrations of a piezo-thermoelastic microbeam, dual-scale nonlocal strain gradient with memory-dependent thermoelasticity, 2025 and stress-gradient piezo-thermoelasticity with nonlocal electron–phonon heat transport, Appl. Math. Model., 2025	The most recent related work; we compare against these to state our novelty
8	Rose, Ultrasonic Guided Waves in Solid Media, Cambridge University Press, 2014	A textbook on Lamb waves and how dispersion is tracked; confirm the edition before citing

Outline of the paper

1. **Introduction:** why plates matter, what size-dependent theories add beyond the classical Eringen model, and the three claims of the paper.
2. **Formulation:** the plate, the field equations for elasticity, piezoelectricity, MGT heat and carriers, the size-effect law, and the boundary conditions with the clear note on extra conditions.
3. **Dispersion relation:** the determinant, and how the modes split into families, depending on the polarisation direction.
4. **Numerical method and verification:** how roots are found and followed, and the table of limiting cases.
5. **Results:** dispersion and complex branches, fading, the thickness surface, length-scale maps, K^2, mode shapes and sensitivity.
6. **Conclusions:** what each claim showed, the limitations, and future work such as surface elasticity, flexoelectricity and graded plates.
7. **Supplementary material:** the code and the full parameter table, so anyone can reproduce the results.

The three claims to keep in view while working:

- A coupled piezo-thermal-semiconductor model of a plate, with a two-length-scale size effect.
- Real and complex branches followed together, with fading and mode behaviour.
- Every step verified against a known limit.